Ali Al-Kelabi



Digital Systems | Embedded Systems | Python & C/C++ | Linux | Hardware-Software Integration

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Germany – Relocation flexible, Ready to move
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PROFIL

Dedicated master's student in the Artificial Intelligence for Smart Sensors and Actuators program at Deggendorf University of Applied Sciences, with a bachelor's degree with honors in mechatronics. Practical experience with Python, C/C++, embedded systems, Linux, Git, Docker and technical problem-solving. Strong interest in digital design, processor architectures, hardware-software integration and systematic verification. I work independently, analytically and with a structured, quality-oriented approach.

English: Fluent in speaking and writing – C2; German: Advanced – (B2 - C1)

EDUCATION

M.Eng. Artificial Intelligence for Smart Sensors and Actuators **March 2025 – Now**
Deggendorf University of Applied Sciences, Germany

- Major Areas: Machine Learning, Deep Learning, Computer Vision, System Design, Edge Devices, Advanced Programming, Big Data
- Current GPA: 2.4

B.Eng. Mechatronics – with Honors **January 2019 – April 2023**
IIUM University Malaysia, Malaysia

- Major Areas: Automation, Mathematics, Robotics, Software Engineering, Machine Learning, Deep Learning, Intelligent Control
- Bachelor's thesis: Research in the field of non-destructive testing and corrosion detection using pulsed eddy current, including signal processing, sensor testing, and 3D probe development.
- GPA: 2.0

WORK EXPERIENCE

Student Assistant – UiPath & C# Automation of Computer Tasks **June 2025 – November 2025**
Deggendorf University of Applied Sciences, Germany

- Development of a UiPath automation course featuring real-world use cases.
- Implementation of web scraping solutions for automated data pipelines.
- Support for process optimization through automation and structured technical implementation.

Project Associate, Deep Learning, Machine Learning, and Computer Vision **March 2024 – July 2024**
ICON, Malaysia

- Contributed to a machine learning project (action recognition) and deep learning (YOLO) to detect specific violations in public spaces, with relevant information—including license plate and route data—transferred to database structures for follow-up.
- Collaboration on a second use case (OCR) for identifying locomotives, extracting visible information and size-related features for tracking purposes.
- Support for Python-based training and data preparation processes, as well as the use of labelling tools such as CVAT and Labellmag.

Work-Study Student in Machine Learning & Sensor Data Analysis **November 2019 – February 2023**
IIUM University Malaysia, in collaboration with the Malaysian Nuclear Agency, Malaysia

- Development and adaptation of 3D designs for PEC probes in the context of non-destructive material testing.
- Conducting pulsed eddy current tests on ferromagnetic test plates using an electrical measurement chain and LabVIEW integration.
- Analysis of the recorded signals to detect and evaluate corrosion under insulation.
- Collaboration on research projects related to signal processing and material inspection.

PROJECTS

Industrial Telemetry and Control System| Personal Project – Electronics, Data analysis and Communication
Python, Raspberry Pi, SQL, LoRa, Edge Computing

- Developed a distributed Raspberry Pi-based sensor-and-control system using Python, Ubuntu and edge computing.
- Measured UV intensity using an RFM95W at 868 MHz and controlled roller blinds and an LED through rule-based logic.
- Built a touchscreen GUI displaying sensor data, graphs, system status, UV sensitivity and alerts.
- Connected Python applications with relational databases and developed monitoring and operations interfaces.
- Embedded systems, sensor integration, real-time system behaviour and hardware-software interaction.

Gesture-Controlled UR5e Pick-and-Place System | Personal Project – Robotics and Machine Vision
ROS 2, Gazebo, MoveIt 2, MediaPipe, OpenCV, Docker

- Developed a ROS 2/Gazebo simulation integrating a UR5e robot, RGB-D camera, MoveIt 2, gesture recognition and a browser dashboard.
- Implemented camera-based object detection and used depth information for 3D point estimation and image-to-world coordinate mapping.
- Developed a click-to-point control feature that converts a camera-image click into a 3D target and validates reachability with collision-checked inverse kinematics.
- Containerized the environment with Docker and created reproducible testing, diagnostics and benchmarking workflows.
- Systematic debugging, software-hardware integration and structured validation.

YOLOv8 Object Detection with ROS 2 and Gazebo | Personal Project – Deep Learning & Robotics
ROS 2, Gazebo, YOLOv8, Python, Computer Vision

- Integrated a YOLOv8 inference pipeline into a ROS 2/Gazebo environment.
- Processed camera images and provided object detections as structured ROS 2 messages.
- Analysed the connection between camera streams, model inference and downstream robot control.
- Real-time data flow, embedded-system thinking and software integration.

Diabetic Retinopathy Detection | Case Study – Deep Learning & Healthcare
Python, TensorFlow/Keras, OpenCV, ResNet-50, DenseNet-121

- Developed a deep-learning pipeline for automated classification and grading of diabetic retinopathy using retinal fundus images.
- Applied image preprocessing, data augmentation, model training, evaluation and architecture comparison.
- Evaluated models using Quadratic Weighted Kappa, macro recall, macro F1 and accuracy.
- Analytical thinking, structured experimentation and systematic evaluation.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, C#, JavaScript, SQL, Git, Docker and Linux/Ubuntu.
- **Digital and Embedded Systems:** Microcontrollers, Raspberry Pi, sensor interfaces, embedded Linux, edge computing and hardware-software integration.
- **Hardware and Systems:** Digital logic concepts, sensors, signal flow, timing behaviour, system integration and technical troubleshooting.
- **Software Development:** FastAPI, Streamlit, ROS 2, Gazebo, technical documentation and reproducible development environments.
- **Data and Analysis:** Pandas, NumPy, SQL, time-series data, anomaly detection and structured evaluation.
- **Tools:** Git, Docker, Linux, VS Code, Jupyter Notebook and technical documentation.
- **Soft Skills:** Strong analytical skills, excellent problem-solving abilities, communication skills, ability to work in a team and collaborate across disciplines, a goal-oriented, structured, and independent approach to work, quick comprehension, and a strong willingness to learn.

07.11.2026

